

Francisco Correia de Sousa

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SUMMARY

MSc in Computer Science and Engineering focused on **distributed systems**, **network security** and **fault-tolerant protocols**. Recent research in asynchronous consensus mechanisms.

WORK EXPERIENCE

Teaching Assistant - IST Computer Science Department Nov. 2024 - Feb. 2026

- Taught lab sessions of **Distributed Systems** (Spring 2025) and **Operating Systems** (Fall 2024 & Fall 2025). Guided students through concurrency models, synchronization, OS fundamentals and distributed protocols.
- Graded and provided feedback of 50+ assignments per semester.

Research Intern - INESC-ID June 2024 - Nov. 2025

- Designed **Chia-Async**, an asynchronous protocol variant removing synchrony assumptions from Chia blockchain consensus via an **Asynchronous DRB** (Distributed Random Beacon).
- Evaluated protocol performance under regular and adversarial conditions: measured latency, throughput, and energy efficiency across Chia and Chia-Async.

PROJECTS

HDS Serenity [Link to Demo](#)

Byzantine fault-tolerant permissioned blockchain implementing Istanbul BFT consensus. Designed test harness covering Byzantine attack scenarios (replay attacks, malicious leaders, quorum failures). Built in Java.

TheCork [Link to Demo](#)

Full-stack encrypted restaurant booking platform with high security guarantees. Architected network topology with firewall, web server, backend, encrypted database and end-to-end TLS encryption. Tech stack: Flask, React, Nginx, MariaDB. Emphasis on secure authentication and encrypted data persistence.

EDUCATION

Instituto Superior Técnico - Computer Science Department Sep. 2022 - Nov. 2025

Computer Science and Engineering **MSc.** GPA: **18/20**

Thesis: "Chia-Async: Towards Asynchronous Proof-of-Space" Grade: **19/20**

Instituto Superior Técnico - Computer Science Department Sep. 2019 - June 2022

Computer Science and Engineering **BSc.** GPA: **15/20**

SKILLS

Technical Areas	Distributed Systems, Network Security, Low-level programming
Programming	C, C++, Python, Java, Kotlin, JS/TS, C#, Go, Rust, SQL
Tools & Systems	Linux, Git, Docker
Language Fluency	Portuguese (Mother Tongue), English (Full Working Proficiency)